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REVIEW

Extraction of lignin and therapeutic applications of lignin-derived compounds. A review

Iuliana Spiridon[1]

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Abstract
Natural polymers are recognized for health improvement and, as a consequence, are widely used in controlled drug delivery, tissue engineering, regenerative medicine, wound dressing, anti-allergy textiles and personal care. Lignin is a valuable, renewable aromatic polymer and a source of low molecular weight chemicals. Lignin can also be converted into biofuels. This article reviews lignin and lignin compounds for medical applications.

Keywords Lignin · Medical applications · Physicochemical and biological properties

Introduction

Lignin is the second most abundant natural polymer of plant biomass. This aromatic three-dimensional macromolecule has an important role in the plant growth, serving as a protective barrier against pathogens (Boerjan et al. 2003; Crini et al. 2019). Also, lignin acts as a natural glue that binds together plant carbohydrates. It is worth mentioning that lignin hinders the access of chemicals and biological agents during delignification and hydrolysis processes (Kumar et al. 2012; Li et al. 2015).

Due to lignin complexity, as well as to unknown associations with other cell wall polymers, lignin macromolecular structure remains elusive. It consists of three basic phenylpropanoic monomers, known as monolignols: *p*-coumaryl, coniferyl and sinapyl alcohols, which are randomly joined to form a three-dimensional aromatic optically inactive amorphous macromolecular polymer.

The main linkages between its monomer units are *β*–O–4′ ether linkages, *α*–O–4′ ether, 4–O–5′ diphenyl ether, *β*–5 phenylcoumaran, *β*–*β*′ resinol, 5–5′ biphenyl and *β*–1′ diphenyl methane (Fig. 1) linkages, their relative abundance being presented in Table 1.

The amount of monolignols, as well as the differences between the linkages present in the macromolecular structure of lignin (Fig. 2), depends on the lignocellulose source (taxa and species, some growing parameters, region) and the extraction methodology, which also influence the amount of hydroxyl, phenolic, carboxyl and sulfonate functional groups in lignin.

Lignin chemofunctionality is provided by the presence of phenolic and aliphatic hydroxyl groups which allow lignin modification by using molecules of interest to medicine, such as peptides or antibodies, as well as the preparation of copolymers as delivery systems for different biomolecules (Spiridon, 2018). Rinaldi et al. (2016) highlighted the importance of the incorporation of new chemically linkages into the lignin in order to open up new perspectives for lignin valorization in the polymer, chemical and fuel industries. Since the market driven view is changing due to the interest in green chemistry, the authors stipulated that an early-stage catalytic conversion of lignin constitutes the backbone of emerging technologies for lignin valorization. The achievements associated with potential applications evidence biological activities of some technical lignins (Espinoza-Acosta et al. 2016), as well as the protective role of lignins against different diseases (Vinardell and Mitjans 2017) or to minimize herbicides effects on the environment (Diyanat and Saeidian 2019). Also, the current applications of lignin hydrogels as stimuli-responsive materials (Rico-García et al. 2020), as well as promising features of lignin as carriers of bioactive molecules (Sipponen et al. 2019), were reported.

✉ Iuliana Spiridon
iulianaspiridon@yahoo.com

[1] Petru Poni Institute of Macromolecular Chemistry, Iasi, Romania

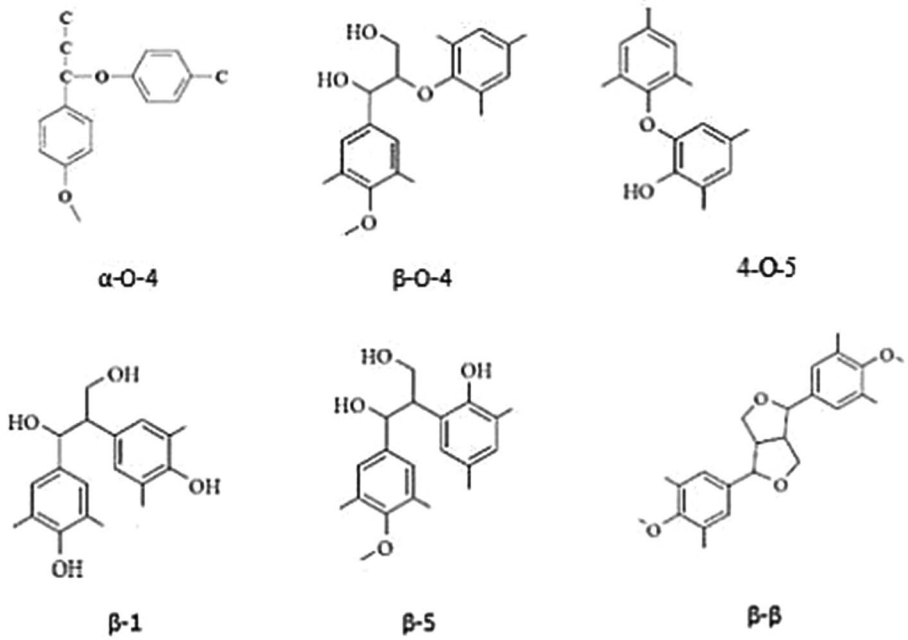


Fig. 1 The main linkages between lignin monomers. The frequency of these linkages has significant impact on the lignin reactivity, being very important to have a good understanding on the relative strengths of these inter-unit linkages in order to separate lignin from different biomass sources

Table 1 The main lignin inter-linkages in biomass (percent of total linkages)

Kind of linkage	Annual plants	Hardwood	Softwood	Reference
α-O-4	Ns	4–8	6–8	Mei et al. (2019)
β–O–4	74–84	60–62	45–50	Guadix-Montero and Sankar (2018)
4–O–5	Ns	3–7	2–5	Singh (2019)
α–O–*γ*′	Ns	2	2	Guadix-Montero and Sankar (2018)
β–1	3	2.6–7	3.5–10	Del Rio et al. (2012), Sette et al. (2011)
β–5	1.8–12.6	9–12	11	Laskar et al. (2013)
β–*β*	2.9–3.8	3	3	Azadi et al. (2013), Herbaut et al. (2018)
5–5′	Ns	6-8	9.5–11	Chen and Wan (2017)

Ns not specified

Extraction

Biomass recalcitrance is related to the presence of lignin, which constitutes a valuable source of high-value products (Fig. 3) such as fuels, chemicals, bioplastics with improved properties. Usually, lignin is generated as a by-product in high amounts in pulp and paper mills and could have multiple value-added applications due to lignin environmental friendliness, low cost, as well as biological properties. It is worth mentioning that the elucidation of the supramolecular structure of lignin and of aspects regarding the possibilities to control lignin chemistry will open up new ways for the synthesis, design and development of novel green products.

Various treatments have been developed in order to remove lignin from lignocellulosic biomass. Many exhaustive reviews (Wang et al. 2019a, b; Seidl and Goulart 2016; Kim et al. 2016; Pan and Saddler 2013) have been published on the main delignification processes, pointing their significant advantages, as well as major bottlenecks/limitations. Therefore, this paper will not reiterate conventional processes for lignin separation.

The most efficient physicochemical delignification methods require large energy consumption and have a harmful impact on the environment. In order to improve the rate of delignification, supercritical fluid technology (Akalin et al. 2017) or physicochemical pretreatments are combined with biological agents (Shirkavand et al. 2016), as well as with microwave and ultrasound irradiation (Singh et al. 2016).

In the last few years, research on the use of ionic liquids in biomass processing has demonstrated that they could offer a sustainable alternative for future biomass refinery processes (Lee et al. 2009; Bodirlau et al. 2010; Halder et al. 2019). However, although the treatment with ionic liquids can improve the efficiency of the delignification process,

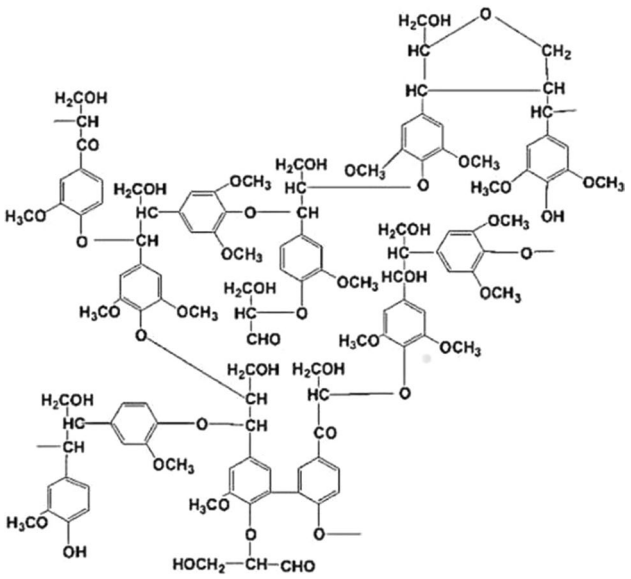


Fig. 2 Partial structure of hardwood lignin. Numerous variations such as the syringyl/guaiacyl ratio, frequency of the linkages between lignin monomers, as well as the degree of condensation occur among the hardwood species

concerns regarding their toxicity and biodegradability, as well as the high operating costs, limit their application in industrial operation.

Lignin biomedical applications

In spite of the complex structure of lignin, lignin posses manifold reaction sites, high content of carbon and low oxygen content, which make it a potential sustainable alternative source of energy and chemicals. In the same time, the circular bioeconomy concept has increased the interest of the scientific community toward lignin exploiting. Being a natural compound, the eco-toxicity of lignin is not a problem and the prospects of lignin medical application have been expanding dramatically in recent years due to the health-enhancing potential (Figueiredo et al. 2018).

The existing literature highlights the high potential of this macromolecular compound and lignin derivatives to be used in the development of novel drugs with different anti-inflammatory, osteoprotective, cardioprotective, neuroprotective and immunoprotective properties (Chowdhury 2014; Kai et al. 2016a, b; Bjørsvik and Liguori 2002; Lee et al. 2012).

Herein, some of the recent advances related to the potential therapeutic applications of lignin and related compounds are presented. Three 4-hydroxycinnamic acid monomers, named caffeic acid, ferulic acid and sinapic acid, were used to synthesize low molecular lignins, which were then modified to introduce sulfate groups in their structure. Both unsulfated and sulfated lignins were tested as inhibitor agents against elastase, oxidation and inflammation (Saluja et al. 2013). The authors reported a triple effect in animal models of experimental emphysema. An alkali-extracted solid residue of hydrolysis lignin alone and in combination with iodine as an enterosorbent PolyphepanTM is used both as a medicine and veterinary compound (Rabinovich 2010).

Lignin in wound-healing dressings

A wound dressing must be able to maintain moisture at the wound surface and to protect the wound site from injury and infection. Also, the exudates absorption as well as the wound pain diminution is required, while the presence of wound-healing agents is welcomed. The development of lignin-based formulations for skin wound healing could become an important topic on the market products for skin replacement.

The arginine surface-modified lignin nanofiber gel was investigated in order to evaluate the possible application as hydrocolloid wound dressing. Due to the structural similarity between extracellular matrix proteins and lignin nanofibers,

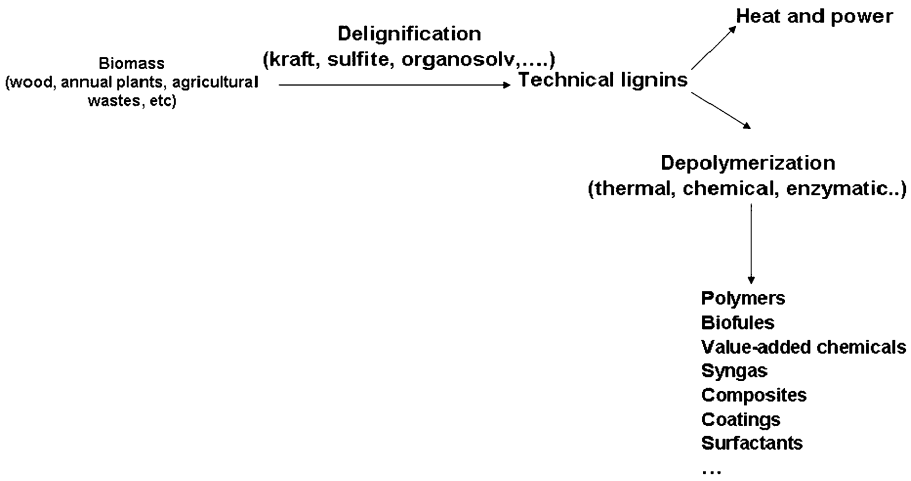


Fig. 3 Biomass processing and possible applications. The lignin recalcitrance resulted into considerable efforts dedicated to the disruption of the linkages in lignin. Since technical lignins are available in huge amounts, the interest for their valorization to generate value-added chemicals, composites and fuels is highly explored

as well as to the wound-healing effects of arginine, the gel was able to promote wound closure, re-epithelialization, angiogenesis and collagen deposition (Reesi et al. 2018; Thakur and Thakur 2015).

A lignin model compound dehydrogenate polymer from coniferyl alcohol—a lignin precursor—was added into alginate gel, and a biomaterial with strong antimicrobial activity against some bacterial strains isolated from patients with chronic wounds was obtained (Spasojevic et al. 2016). The efficiency of the biomaterial was in the same range as that of streptomycin and ampicillin antibiotics, with no toxic effect on human epithelial cells.

A recent study (Dominguez-Robles et al. 2019) examined 3D printing meshes, comprising poly(lactic acid), lignin and curcumin, which are efficient in treating chronic wounds. The paper highlighted the possibility to valorize the antioxidant capacity of lignin by developing wound dressing materials. Figueiredo et al. (2017) reported the adsorption of β-casein on the surface of lignin. Lignin nanoparticles were modified with positively charged poly(L-lysine) and then treated with negatively charged poly(L-glutamic acid) in order to increase the reactivity between nanocellulose and the protein-coated lignin particles. After agglutination and analysis of chamois specimens, the potential of colloidal lignin particles in wound sealing was demonstrated.

The electrospinning technique was used to develop chitin–lignin poly (glycerol sebacate) hybrid fibers (Abudula et al. 2018) with improved antibacterial performance due to the incorporation of poly (glycerol sebacate) and potential application in wound care products. In another study, lignin was used as ionotropic cross-linker for chitosan to obtain a hydrogel (Ravishankar et al. 2019) with wound-healing potential.

Antiviral effect of lignin and lignin compounds

Lignosulfonic acid, from sulfite delignification of wood, has been explored in terms of antiviral activity against human immunodeficiency virus (HIV) and genital herpes simplex (HSV), respectively (Qiu et al. 2012; Gordts et al. 2015). Also, low molecular organosolv lignin was able to inhibit the replication of human immunodeficiency virus type 1 by suppression of the genital herpes simplex type 1 (HIV-1) transcription mechanism (Mitsuhashi et al. 2008). Another report (Gonzalez Lopez et al. 2009) revealed that oral uptake of pine cone lignin–ascorbic acid tablets resulted in shortening the outbreaks and in relieving the symptoms of HSV-1-infected patients, thus promoting faster healing. Also, the antiviral and the immunostimulatory effect of lignin–carbohydrate–protein complexes have been demonstrated (Lee et al. 2011; Zhang et al. 2007). The same effects have been reported for the lignin–carbohydrate complex separated by alkaline extraction from cacao (Sakagami et al. 2011). The higher anti-HIV activity, as compared to that of lipopolysaccharides, has been pointed out.

Hardwood lignin and ursolic acid (isolated from *Salvia officinalis*) were assessed on hepatoma cells (HepG2) and colonic cells (Caco-2) as protection agents against the genotoxic effect of 3′-azido-3′-dideoxythymidine, an anti-HIV drug. The report (Slameňová et al. 2006) evidenced their positive influence against the deoxyribonucleic acid-damaging effects of the drug. Other lignins separated from some species of Cactaceae (José da Cruz et al. 2019) were assimilated as molecules with a proinflammatory profile, being able to promote immune stimulation through cell activation and proliferation, as well as by the production of cytokines.

Lignin in diabetes treatment

Diabetes is a major public health threat and requires daily self-care. It could reduce life expectancy when different related health complications occur. It is important to develop new therapeutic agents to optimize the quality of life or to control the evolution of the disease. Some authors (Barapatre et al. 2015) reported the inhibitory effect of modified lignin on α-amylase, as well as on *E. nidulans* and *A. flavus*, respectively. The study has evidenced the potential antihyperglycemic properties of the compound, which are due to the limitation of postprandial glucose absorption. The new hydrogen bonding between lignin and enzymes resulted in the inhibition of the enzyme activity. Similar results were obtained by use of lignin isolated from oil palm empty fruit bunches (Medina et al. 2016).

Hasegawa et al. (2017) reported the inhibition of mitotic clonal expansion of preadipocyte cells, as well as of glucose uptake into adipocytes after treatment with lignosulfonic acid. The effect of lignosulfonic acid on the intestinal epithelial barrier function was studied by other authors (Matsuhisa et al. 2018) who found that the barrier function of tight junctions was improved. Moreover, the tight junction barrier integrity impaired by inflammatory cytokines was restored.

Nordihydroguaiaretic acid, a lignin compound separated from *Larrea divarita* shrubs (Chopra 2004), was found efficient in solving problems related to rats' diabetic nephropathy. The chronic treatment with nordihydroguaiaretic acid was recorded to lead to a high prevention rate of both renal dysfunction and oxidative stress. The authors stipulated that the antioxidative properties of nordihydroguaiaretic acid may involve xanthine oxidase inhibition, iron chelation, as well as the direct scavenging of free radicals. The capacity of nordihydroguaiaretic acid to reduce cytokine and nitric oxide production (Hsu et al. 2001) was also reported.

Lignin sorbents with hypolipidemic and hypoglycemic effects have been obtained by condensation of lignin with epichlorohydrin oligomer and amination of the formed

chloroderivative with polyamines (Chopabayeva 2017). They could be useful to diabetic retinopathy patients as they contribute to the removal of water and lipid soluble toxic metabolites from blood serum.

Hydroxytyrosol, a phenol separated from fresh olive mill waste, exhibited hypoglycemic and antioxidant effects on diabetic rats, with a positive influence on pancreatic β-cells (Hamden et al. 2009). Chlorogenic acid, a phenolic compound, was efficient in the diminution of the glucose level in type 2 diabetes by lowering the postprandial hyperglycemic peak (Bassoli et al. 2009).

The effect of lignophenols from beech (Sato et al. 2009) was explored in the diabetic nephropathy of rats. It was found that the treatment with lignophenols diminished the oxidative stress. Also, streptozotocin-induced diabetic rats exhibited a positive evolution due to the superoxide (O_2^-) reduction in the renal cortices, as well as due to the activation and infiltration of macrophages in the glomerular and tubulointerstitial areas. Other authors (Norikura et al. 2010) reported on the antioxidant and anti-diabetic properties of lignophenols extracted from beech and established that they reduced the secretion of apolipoprotein B and cholesterol levels in the human hepatocellular carcinoma cell line.

Lignin-based materials as delivery systems

Numerous researchers are focused to design biobased nanocarriers, where active bio agent can be encapsulated or physically attached. Drug delivery systems with better targeting mechanisms could improve the efficacy of therapeutics by increasing bioavailability of a drug locally rather than systemically. Spherical biocarriers using lignin–carbohydrate complexes from ginkgo (*Ginkgo biloba* L.) were prepared using liquid nitrogen freeze-drying method and used to culture of human hepatocytes (Fig. 4). The study demonstrated that spherical biocarriers based on lignin–carbohydrate complexes could be used as biomaterials in the tissue engineering of culture hepatocytes to create hepatic organs (Zhao et al. 2017).

Human cervical adenocarcinoma cells have been used to evaluate the cytotoxicity of different kinds of lignin. The tests have revealed the binding capacity of DNA (deoxyribonucleic acid) to lignin nanotubes, thus demonstrating that lignin can be turned into a potential vehicle of DNA in living organisms (Ten et al. 2014). An important conclusion of this study was that lignin-isolation procedure was mainly influencing the cytotoxicity. Also, the transfection efficiency was influenced by the lignin source.

Figueiredo et al. (2018) have studied the possible use of Fe_3O_4-lignin nanoparticles in the therapy and diagnosis of cancer. They found that Fe_3O_4-lignin nanoparticles become magnetized up to their saturation magnetization by the application of a magnetic field. Thus, when using a magnetic field, they acted as a vehicle for the delivery of drug molecules to a specific target in the body. At the same time, it was demonstrated that lignin nanoparticles can serve as carrier for poorly water-soluble drugs (benzazulene and sorafenib), as well as for a water-soluble anticancer drug (capecitabine), which have inhibitory effects on different cancer cells. Recently, Dai et al. (2017) have reported that lignin could be used as carrier for resveratrol, which was encapsulated with Fe_3O_4 nanoparticles for potential use in the therapy of cancer. Biocompatibility and tumor suppression tests did not evidence any adverse effects on cells nor on mice. Moreover, lignin nanoparticles presented a high drug-loading capacity of over 20 wt% as well as reduced adverse effects compared to resveratrol.

Coumarin-6 was ultrasonically loaded into kraft lignin, and all the microcapsules were internalized into cytoplasm ovary cells (Tortora et al. 2014). The study evidenced no

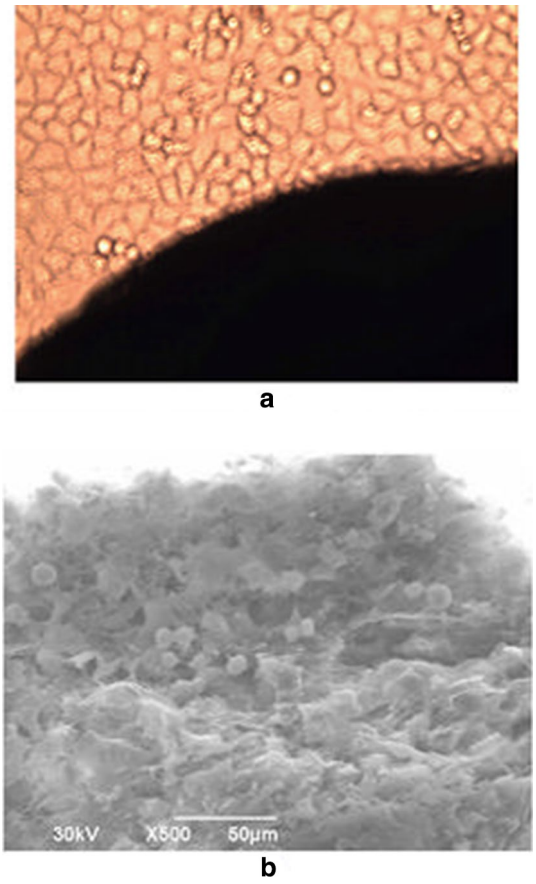


Fig. 4 An inverted microscope image (a) and SEM image (b) of the human hepatocytes L-02 cultured on the spherical biocarriers. Human hepatocytes (L-02) were cultured statically in the presence of biocarriers. They adhered to the spherical biocarriers, indicating that spherical biocarriers based on the lignin–carbohydrate complex is biocompatible and suitable for the hepatocyte culture. SEM: scanning electron microscopy. Reproduced from Zhao et al. 2017, available via license Creative Commons Attribution

cytotoxic effect of the lignin microcapsules and their possible biomedical applications. The release experiment was performed in water using the same surfactant and surfactant concentration as used by Chen et al. (2016).

The efficiency of a novel folic acid-polyethylene glycol-lignin as nanocarrier for anticancer drug (10-hydroxycamptothecin) was demonstrated (Liu et al. 2018). The tests evidenced that the drug preferentially accumulated at the tumor site, protecting the normal organs.

Li et al. (2017) developed some lignin micelle formulations for delivering ibuprofen. The formulations could be used in the treatment of some painful conditions and musculoskeletal disorders. Lignin was esterified with poly(ethyleneglycol) and poly(methyl vinyl ether-co-maleic acid) to obtain a hydrogels able to deliver curcumin (Larraneta et al. 2018). The antibacterial properties against *Staphylococcus aureus* and *Proteus mirabilis* recommend the studied material as a medical coating material (Fig. 5).

Sugarcane lignin was able to deliver methotrexate, a drug utilized in the treatment of rheumatoid arthritis (Wahba et al. 2015). Due to the good dispersion in the intercapillary blood vessels, lignin was able to carry the drug toward inflamed tissues. Also, kraft lignin was used to develop materials for gene delivery (Liu et al. 2015). The obtained copolymers have proved DNA-binding capability and non-cytotoxicity, while the gene transfection efficiency was influenced by the chain length of the lignin-copolymers.

A recent study reported the use of the lignin as excipient for tablet manufacturing (Domínguez-Robles et al. 2019), that is not an ideal excipient for tablet manufacturing when is used in high concentration. The use of low concentration (20%) of lignin (Pishnamazi et al. 2018) or modified lignin (10%) as an excipient in oral dosage forms (Pishnamazi et al. 2019) for controlled-release of different drugs proved that lignin acts as a tablet binder.

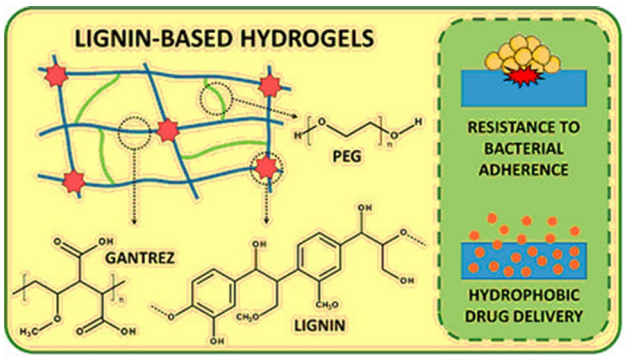


Fig. 5 Lignin-hydrogel as a medical coating material. The hydrogels were obtained by esterification reaction, combining lignin with poly(ethylene glycol) and poly(methyl vinyl ether-*co*-maleic acid). After loading of cucurmin, the hydrogels were able to sustain the delivery of this compound for up to 4 days. Also, significant resistance to bacterial adherence when compared to poly(vinyl chloride) was demonstrated. Reproduced from Larraneta et al. 2018, with permission from ACS

Lignin in anticancer applications

Cancer is a growing health problem, being one of the most important mortality factors. The problems related to the standard treatments, such as the emergence of resistance to anticancer drugs and the high toxicity levels, represent limitations of the current standard treatments. On the other hand, the use of natural compounds to reverse or suppress the initial phases of carcinogenesis, as well as the development of new treatment schemes including biomass compounds is a challenge for the scientific and medical world.

The cytotoxic effect of lignin isolated from bagasse, as well as of lignosulfonates, was investigated by Ugartondo et al. (Ugartondo et al. 2008) using human keratinocyte HaCaT and murine fibroblast 3T3 cells. These compounds exhibited antioxidant effects at non-cytotoxic concentrations, while a more cytotoxic effect on the keratinocyte cell line, as compared to murine fibroblast cells (but only at very high concentrations), was evidenced.

Some authors (Huang et al. 2018a, b) reported that lignin–carbohydrate complexes from bamboo may reduce the incidence of breast tumor. Immunological analysis evidenced both growth inhibition in breast tumor cells and an immunostimulatory effect on macrophage cells. According to Sakagami et al. (2005), a lignin–carbohydrate complex comprising 20% carbohydrate also exhibited antitumor activity. The authors highlighted the importance of sugar and polyphenol conjugation, stating that the induction of antiviral activity is due to the polymerized structure of phenylpropenoids from the lignin–carbohydrate complexes, while the immunopotentiating effect is due to the higher-order structure of lignin and polysaccharides.

Anticoagulant effect of lignin

Thrombotic disorders are usually treated using anticoagulants, such as heparin and warfarin, in spite of serious bleeding risks and adverse reactions. That is why, the interest in natural compounds with both anticoagulant and antiplatelet activities for potential applications in thrombotic events is growing. Sulfated low molecular weight lignins designed as oligomeric mimetics of the low molecular weight heparins by chemoenzymatical methods exhibited an anticoagulation potential, as mentioned by Abdel Aziz et al. (2011). Thus, sulfated low molecular lignins (Henry and Desai 2014; Henry et al. 2012) can inhibit the catalytic activity of thrombin and factor Xa (a specific protease used in the blood clotting system). It seems that these lignins induce allosteric inhibition of some coagulation proteases by using the heparin binding domain. Also, it was proved that sulfated

β–O4 lignins act as allosteric inhibitors of thrombin (Mehta et al. 2016). As compared to heparin, the lower charge density of sulfate groups tends to induce binding to a few blood proteins and cells, which may reduce bleeding. Recently, the anticoagulant effect of lignin-capped silver nanoparticles (LCSN) (Marulasiddeshwara et al. 2017) synthesized under green reaction conditions was tested. It was evidenced that it did not alter plasma clotting time up to the tested concentration of 30 µg, suggesting that lignin-capped silver nanoparticles could be efficient in the treatment of thrombotic disorders.

A polyphenolic–polysaccharide complex isolated from the aerial parts of *Agrimonia eupatoria* (Tsirigotis-Maniecka et al. 2019), as well as that separated from *Fragaria vesca* (Pawlaczyk-Graja et al. 2019), exhibited a high selective anticoagulant activity.

Lignin-based scaffolds for tissue engineering

A recent review (Witzler et al. 2018) evidenced that hybrid lignin materials could be used as scaffolds for bone regeneration in stem cell-based therapies. Other applications include prosthetics or implantable devices. Spiridon and Tanase (2018) obtained some polylactic acid–lignin composites, with no cytotoxic effect on SaOS-2 cells. Also, glycinated Kraft lignin (Musilová et al. 2018) obtained after the Mannich reaction was cross-linked to hyaluronan and the obtained hydrogels were found suitable for different biomedical applications, such as tissue engineering due to their mechanical properties, stability, swelling capacity and viscoelastic behavior, as well as owing to their cytotoxicity effect.

Quraishi et al. (2015) have developed an alginate-lignin aerogel material. The cell adhesion tests have revealed the ability of mouse fibroblast-like L929 cells to adhere onto the surface of the alginate-lignin materials. Also, by electrospinning of poly(L-lactide) (PLLA), poly(lactic) acid (PLA) and lignin, new materials with antioxidant activities and biocompatibility for rat neural stem (PC12) cells, human dermal fibroblasts and human mesenchymal stem cells (MSCs) were obtained by Kai et al. (2016a, b). The experimental results recommend them to be exploited as tissue engineering scaffolds to attenuate cellular oxidative stress. The same authors used lignin to develop lignin-poly(3-hydroxybutyrate) composite nanofibers (Kai et al. 2019) with potential for tissue engineering. Polycaprolactone/lignin nanofibers (Wang et al. 2018a, b) were investigated as a material for nerve tissue repair and regeneration. The cell proliferation of both Schwann cells and bone marrow mesenchymal stem cells, as well as the stimulation of neurite outgrowth of dorsal root ganglion neurons, proved that polycaprolactone/lignin scaffolds could be a promising substrate for nerve regeneration and functional recovery in peripheral neural tissue engineering.

Alkali lignin and organosolv lignin were tested as inhibitors of the tyrosinase activity (Wang et al. 2018a, b, 2019a, b), and the anti-tyrosinase activity of organosolv lignin was noted to be higher than that of alkali lignin.

Recently, the radioprotective effect of a lignin-derived polyphenolic formulation, comprising ammonium molybdate, in animals exposed to a 5.0–8.0 Gy dose (Bykov et al. 2018) has been reported. It seems that the enhancement of extramedullary hematopoiesis in the spleen and the protective effect on the intestinal epithelium are responsible for the action of the studied compound against the radiation effects. Other works mentioned that the chronic use of the lignin complex had no adverse effects on the hematopoietic system (Prasad 2005).

Antioxidant activity

Lignin and lignin compounds exert antioxidant activity (Dizhbite et al. 2004; Vinardell et al. 2008; Košíková and Lába 2009; Huang et al. 2018a, b; Xu et al. 2018; Farooq et al. 2018) due to their polyphenolic structure.

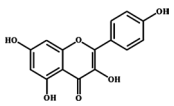
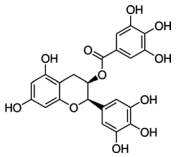
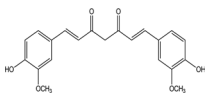
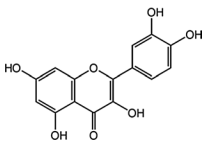
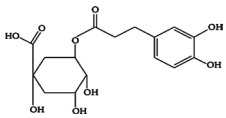
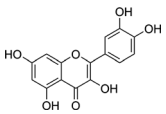
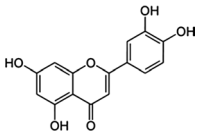
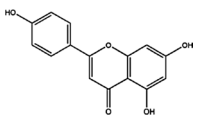
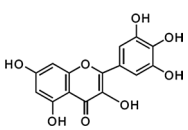
The addition of 1–3 wt% lignin in polyvinyl alcohol/chitosan hydrogels demonstrated that the presence of lignin improved both the antibacterial and the antioxidant properties of the material (Yang et al. 2018). After infusion of indulin AT™ lignin in an aqueous solution of silver nitrate ($AgNO_3$), followed by coating with polydiallyldimethylammonium chloride, the lignin particles were able to neutralize common Gram-negative and Gram-positive human pathogens (Richter et al. 2015). Gregorova et al. (2011) confirmed the great potential of Björkman lignin isolated from beech as antibacterial agent against *E. coli* and *S. aureus*. This kind of lignin exhibited an antimicrobial effect comparable with that of bronopol and chlorhexidine.

Ryu and Na (2018) have evaluated the antioxidant effect of syringic acid and evidenced the possibility to be used as natural phytochemical in cosmetic products. They proved that syringic acid had decreased the production of collagen-inhibiting matrix metalloproteinase-1 through the activator protein-1 mechanism and; as a result, human keratinocyte cells were protected against ultraviolet B damage. A similar effect was reported for hydroxytyrosol (D'Angelo et al. 2005) and for lignin–carbohydrate complex from *Sasa senanesis* Rehder (Sakagami et al. 2016). Qian et al. (2017) pointed that the addition of colloidal lignin spheres into skin sunscreen cream had boosted UV-blocking properties.

Phenolic compounds such flavonoids (anthocyanidins and anthoxanthins) or diferuloylmethanes, stilbenes and phenolic acids present biological activity which recommend them as up-and-coming compounds for the developments of new natural therapies for different diseases.

As can be seen in Table 2, polyphenolics exhibit antibacterial, anti-inflammatory, antiallergic, hepatoprotective, antithrombotic, anticancer and antiviral activities (Treutter 2006; Ciubara et al. 2018; Spiridon et al. 2011; Velde et al. 2019; Amatori et al. 2016; Xiang et al. 2019), thus supporting a multitude of medical applications of these natural compounds. The antioxidant capacity is considered as a first-step defense against the mutagenesis process. The capacity of polyphenols to scavenge reactive oxygen species led to the decrease in DNA oxidative damage, to stimulate antioxidant enzymes and thus to the enhancing of DNA repair (Amatori et al. 2016). For example, catechins have numerous physiological functions, a schematic diagram of metabolism of green tea catechin being presented in Fig. 6. A recent study evidenced that it inhibited the preadipocytes differentiation and lipid decomposition of mature adipocytes (Jiang et al. 2019). Also catechins are able to promote the proliferation of beneficial intestinal

Table 2 Recent medical applications of polyphenols

Compound	Structure	Potential applications	Reference
Kaempferol		Formulations that can candidate as wound healing or anticancer agents; preparations to ameliorate kidney damage induced by diabetes	Kashyap et al. (2017) Thangavel et al. (2018) Sharma et al. (2019) Ozay et al. (2019)
Epigallocatechin gallate		Pro-osteogenic agents for the therapy of osteoporosis, to induce cell apoptosis in different cancer diseases	Jin et al. (2014) Jiang et al. (2018) Vazquez-Calvo et al. (2017)
Curcumin		Curcumin formulations with anticancer or anti-viral potential or to accelerate wound healing	Lee et al.(2014) Panchatcharam et al. (2006) Mathew and Hsu (2018) Stanic (2017) Hodaei et al. (2019) Nautiyal et al. (2011) Farooq et al. (2019) Walia et al. (2019)
Quercetin and aglycones		Formulations with anti-viral, anti-diabetic, hypocholesterolemic, gastroprotective effects	Gomez et al. (2008) Shi et al. (2019) Yarahmadi et al. (2018) Kakkar et al. (2018)
Chlorogenic acid		Treatment or prevention of some diseases (cardiovascular, type 2 diabetes), prevention of neurodegenerative disorders	Naveed et al. (2018) Gouthamchandra et al. (2017) Zhang et al. (2019)
Resveratrol		Booster for immunotherapy, treatment of different autoimmune and inflammatory diseases due to the modulation of both oxidative stress and inflammatory response	Zhao et al. (2018) Zhou et al. (2019) Chen et al. (2019) Hao et al. (2018) Arora and Jaglan (2018)
Luteolin		Prevention or treatment of gastric cancer, formulations to modulate obesity-associated metabolic disorders	Zhou et al. (2018) Aziz et al. (2018) Imran et al. (2019) Ahmed et al. (2019)
Apigenin		Agents for different diseases (glucose metabolism, hepatic steatosis, gene regulation in different carcinogenesis)	Balez et al. (2016) Hong et al. (2018) Zhou et al. (2017)
Myricetin		Supplements to prevent and treat different metabolic diseases	Miyazaki et al. (2018) Akindehin et al. (2018) Chen et al. (2019) Devi et al. (2015)

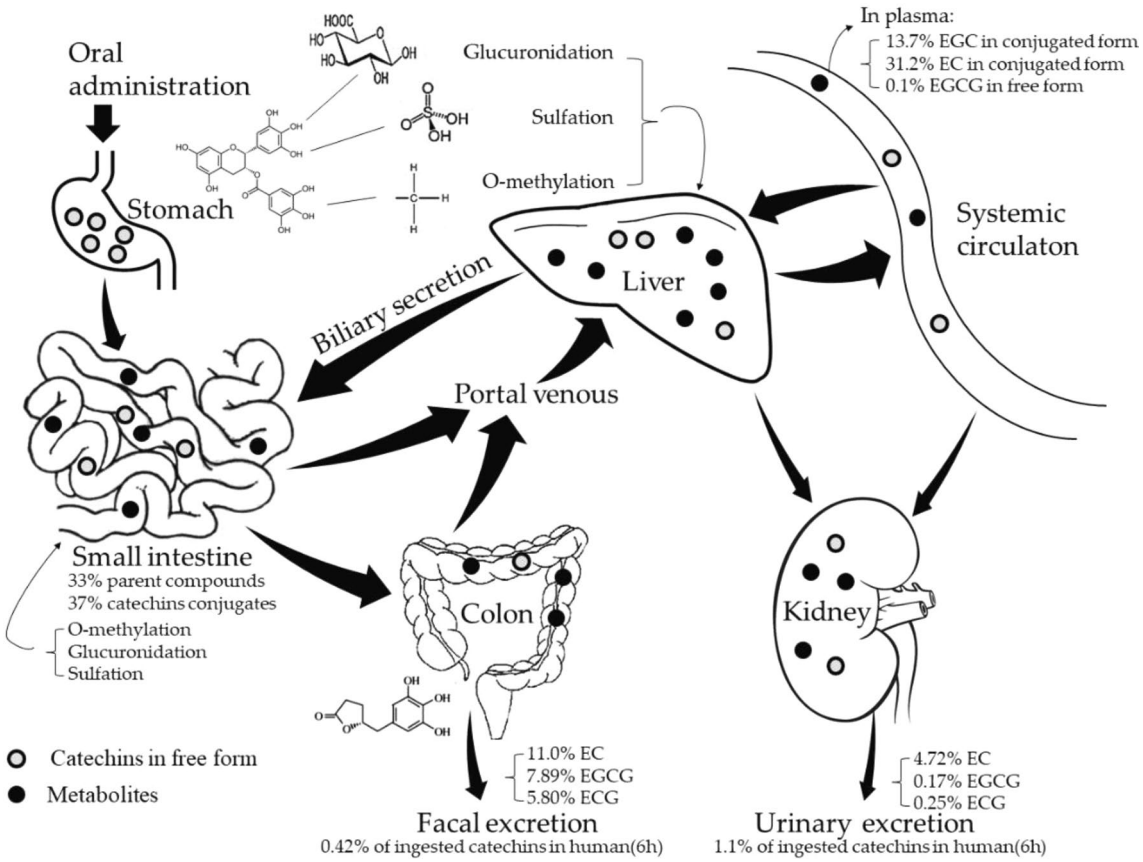


Fig. 6 Metabolism of green tea catechins. Epigallocatechin gallate (ECGC) became active efflux after absorption by adenosine triphosphate-depended proteins. EGCG and its metabolites represent the substrates for multidrug resistance-associated protein (MRP), as well as for P-glycoprotein (P-gp) efflux pumps and further the most absorbed catechins are pumped back to the extracellular or intestinal space. (Reproduced from Cai et al. 2018, MDPI, available via license Creative Commons Attribution)

bacteria and thus regulate the balance of intestinal flora (Fan et al. 2017).

Although lignin has a heterogeneous character and a limited reactivity, new applications as biomaterials, devices for biological detection and remediation in human health seem to be steadily increasing within the requirements in the marketplace.

Conclusion

Lignin has gained huge interest due to the potential use in different fields. This review has surveyed the most recent research on the therapeutic effects of lignin and lignin compounds. Literature data evidenced the antioxidant, antiviral, anticancer, as well as immunomodulatory effects of lignin and lignin compounds. Also, the potential of lignin to be used in alleviating problems related to diabetes and obesity control was reported. Having in mind that lignin could be a sustainable alternative to fossil resources, more investigations dedicated to deeper understanding of the biological activity and intracellular mechanisms are needed to develop materials for potential high-value markets, including medicine, pharmaceuticals and bioplastics, as well as other valuable chemicals. Unfortunately, lignin cannot be approached as homogeneous raw material, having a highly complex structure and different physicochemical properties, as a function of biomass sources and delignification processes. This is a high challenge because a correlation between lignin source, delignification process and biological effect could be useful for the development of new products with therapeutic effects. Moreover, the bioactivities of different kinds of lignin and lignin compounds should be assessed by in vitro, in vivo and cellular assays in order to establish the functional mechanisms. These studies will pave the way for innovative treatment options for different diseases.

Author contributions IS conceived, designed and wrote the paper

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